

IS1102-2 – INTRODUCTION TO DATA SCIENCE
(For IS)

Max. Marks: 100

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

20 Marks

- 1-

12. Which of the following is not a typical outcome of data mining?
 A) Improved decision-making B) Identification of trends
 C) Data encryption D) Prediction of future events
13. What is "data reduction" in data mining?
 A) The process of removing redundant data B) The process of summarizing data
 C) The process of cleaning and transforming data D) The process of storing data in a compressed format
14. Which of the following is a common use of data mining in the telecommunication industry?
 A) Call drop analysis B) Signal loss prediction
 C) Fraud detection D) Hardware maintenance
15. The problem of finding patterns that are valid, novel, useful, and understandable in data mining is known as
 A) Knowledge discovery B) Pattern recognition
 C) Data cleaning D) Pattern validation
16. Principal Component Analysis (PCA) is a technique used in
 A) Data cleaning B) Data transformation
 C) Data reduction D) Data integration
17. Which of the following is a data transformation technique?
 A) Data integration B) Normalization
 C) Data cleaning D) Sampling
18. In supervised learning, the training dataset consists of:
 A) Input features only B) Input features and output label
 C) Output labels only D) No features and no label
19. The main objective of a classification algorithm in supervised learning is to:
 A) Assign input data to predefined categories or classes B) Predict continuous values
 C) Determine the optimal number of clusters D) Identify patterns in unlabeled data
20. What is the role of the validation dataset in machine learning?
 A) To fine-tune the model's hyperparameters B) To train the model
 C) To test the model's performance D) To prevent underfitting

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I

- | | Marks | BT* | CO* | PO* |
|--|-------|-----|-----|-----|
| 1. a) Explain the steps involved in the data science life cycle. | 8 | L*2 | 1 | 1 |
| b) Explain how skewness and Pearson's Median Skewness Coefficient are used to measure data sample asymmetry. | 8 | L2 | 1 | 1 |

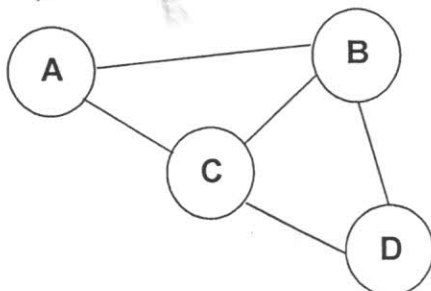


Fig. 2a

- | | | | | |
|---|---|----|---|-----|
| 2. a) Determine the ego network for each node in the graph (Fig. 2a), and which node has the most influential ego network based on centrality measures. | 8 | L3 | 2 | 1,2 |
| b) Outline the key steps in conducting Exploratory Data Analysis (EDA) on a dataset and describe the purpose of each step. | 8 | L2 | 1 | 1 |
| 3. a) Discuss different steps of data preparation. | 6 | L3 | 1 | 1 |

- b) What is the betweenness centrality of each node in the given graph (Figure.3b), and which node has the highest betweenness centrality?

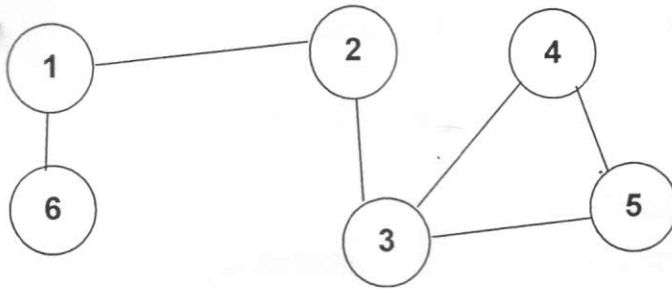


Figure. 3b

10 L3 2 1,2

Unit – II

- | | | | | | |
|-------|---|---|----|---|-----|
| 4. a) | Explain the different challenges of data mining. | 8 | L2 | 3 | 1,2 |
| b) | Discuss the concept of classification and clustering. | 8 | L2 | 3 | 1,2 |
| 5. a) | Explain different measures of data quality. | 8 | L2 | 3 | 1,2 |
| b) | Discuss on how to handle missing data and noisy data. | 8 | L2 | 3 | 1,2 |
| 6. a) | Discuss different data warehouse applications. | 6 | L2 | 4 | 1,2 |
| b) | Explain different data warehouse models. | 6 | L2 | 4 | 1,2 |
| c) | Explain multiple linear regression. | 4 | L2 | 4 | |

Unit – III

- | | | | | | |
|-------|---|----|----|---|-----|
| 7. a) | Discuss the role of training, validation and test data. | 6 | L2 | 5 | 1,2 |
| b) | Considering confusion matrix, explain the different performance metrics of a classifier algorithm. | 10 | L2 | 5 | 1,2 |
| 8. a) | Explain the concept of clustering. Also, discuss different types of inputs to be considered while grouping. | 8 | L2 | 5 | 1,2 |
| b) | Discuss the role of similarity and distances in clustering. | 8 | L2 | 5 | 1,2 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
